

Mark Model v2 — Event-Based Wetland–Flood Analysis with Upgraded Steps 1–3

RREDI event detection · 800 m PRISM rainfall · physical wetland effectiveness W

MMSD Wetland–Flood Effectiveness Study

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Abstract

This report documents a full re-run of the event-based wetland–flood pipeline on the nine long-record MMSD gauges, with the first three stages upgraded and stages 4–6 left byte-for-byte unchanged. Step 1 replaces Eckhardt/peak-prominence event detection with a RREDI-style rolling-median, ratio-threshold detector (6,858 quality-controlled events, 2001–2023). Step 2 replaces the 12 km IEMRE Stage IV rainfall with native 800 m PRISM, eliminating convective smearing over small basins, and adopts a corrected effective-rainfall antecedent index (P_{eff} ; Kohler & Linsley 1951). Step 3 replaces the connectivity-weighted wetland fraction with a physically grounded effectiveness metric $W = \sum_i A_i S_i K(T_{ij})$ (kinematic celerity travel times, wetland roughness, the redundant connectivity term removed, and a type-weighted glaciated storage). Two thin adapters bridge the schema so the unchanged downstream stages run verbatim. Over 5,907 usable gauge-events, the central result is not only reproduced but *strengthened*: under wild cluster bootstrap the standardized peak-shaving interaction $W \cdot P$ on log peak discharge is -0.233 ($p = 0.010$), larger than the -0.155 of the original pipeline. The wetland level effect remains unidentifiable across nine confounded gauges. Everything runs in an isolated `Mark_model_v2/` directory; no existing file is modified.

1 What changed in v2, and what did not

The pipeline is seven stages. Stages 1–3 (the variable-construction stages) were upgraded; stages 4–6 are the unchanged workflow.

Two schema adapters make the unchanged stages run without edits: (i) the wetland adapter writes the v2 metric into the contract file `panel_W_Sw.csv` (treatment $W_{\text{exp}} \leftarrow W_{v2}$; the regression standardizes W , so the scale change is harmless); (ii) the rainfall step exposes P_{eff} under the legacy control name `api_30_mm`.

2 Study area and gauges

The nine long-record USGS gauges span the Milwaukee/Menomonee wetland–urbanization gradient, from large wetland-rich rural rivers (Milwaukee R., Cedar Creek) to small flashy urban creeks (Honey Creek, Underwood Creek). Catchments, routing, travel-time surfaces, and conditioned DEMs are reused read-only from the existing pipeline.

Stage	v2 change	Status
1 Events	Eckhardt/peak-prominence → RREDI rolling-median ratio-threshold detector; reads cached raw 15-min discharge offline	Upgraded (7,355 raw / 6,858 QC-pass)
2 Rain-fall	IEMRE 12 km → 800 m PRISM basin averaging; <i>api_30</i> → corrected P_{eff} antecedent index	Upgraded
3 Wet-land W	connectivity fraction → $W = \sum A_i S_i K(T)$ (celerity T , wetland roughness, C dropped, type×storage S_i)	Upgraded
4 Con-trols	drainage area, slope, impervious, soils	<i>Unchanged (verbatim)</i>
5 Panel	join to gauge×event; collinearity diagnostic	<i>Unchanged (verbatim)</i>
6 Mod-els	nested Models 1–4; MixedLM + clustered SE; placebo, LOGO	<i>Unchanged (verbatim)</i>
7 Boot-strap	exact wild cluster bootstrap ($2^9 = 512$)	<i>Unchanged (verbatim)</i>

3 Step 1 — RREDI event detection

Events are detected on the hourly discharge series by (a) a centered 7-day rolling-median baseline, (b) event start when $Q/\text{baseline} \geq 1.20$, (c) event end after the ratio stays below 1.05 for 6 h, (d) merging events separated by < 12 h, and (e) keeping events with a peak ratio ≥ 1.30 ; quality control screens data coverage (≥ 0.80), gaps (≤ 2 h), and sampling interval. This yields **6,858 QC-pass events** (Table 1), roughly $2.5\times$ the original Eckhardt detector, because the ratio detector resolves many smaller rises. Raw 15-min discharge is read from a local cache (no re-download); gage height is not available in this mode, so peak-stage metrics are absent.

Table 1: Step 1 (RREDI) — QC-pass event counts and medians per gauge, 2001–2023.

Site	QC-pass events	med. dur. (h)	med. TTP (h)	med. Q_p (cfs)
04087000	478	29.8	3.6	789
04086500	353	66.8	18.5	138
04086600	274	55.2	7.6	380
04087030	737	55.0	2.4	60
04087050	506	52.2	6.0	13
04087070	987	44.5	5.8	38
04087120	954	45.7	3.2	408
04087088	1303	43.1	3.6	62
04087119	1266	37.8	3.8	43
Total	6858			

4 Step 2 — 800 m PRISM rainfall

Basin-mean daily rainfall is now the mean of the native 800 m PRISM cells whose centres fall inside each catchment, replacing the 0.125° (≈ 12 km) IEMRE analysis. This removes the single-pixel smearing that previously afflicted the four smallest (< 50 km²) basins: those catchments now contain dozens of independent rainfall cells. The annual basin precipitation (Table 2, 896–964 mm)

validates the averaging against the southeastern-Wisconsin climatology. Event depth P_e , volume V_e , intensity, and the runoff coefficient are computed by direct summation over each event window. The antecedent variable is the effective cumulative rainfall $P_{\text{eff}}(t) = \sum_{n=0}^{30} k^n P(t-n)$ ($k = 0.90$; Kohler & Linsley 1951), with a prior off-by-one lookup error corrected and the storm’s own start-day rainfall included.

Table 2: Step 2 (800 m PRISM) — per-gauge rainfall summary.

Site	area (km ²)	events	ann. precip (mm)	med. P_e (mm)	med. RC	med. P_{eff} (mm)
04087000	1751	478	899	15.5	0.017	25.6
04086500	321	353	918	21.8	0.029	36.1
04086600	1518	274	896	16.2	0.015	31.6
04087120	310	954	917	11.5	0.080	20.5
04087030	85	737	910	13.8	0.030	20.9
04087050	21	506	916	16.8	0.032	23.2
04087070	50	987	910	10.7	0.066	20.4
04087088	47	1303	937	8.7	0.091	18.9
04087119	27	1266	964	6.0	0.175	20.7

Caveat — runoff coefficients. Median runoff coefficients (0.02–0.18) are lower than in the original pipeline (0.16–0.32). This is a mechanical consequence of the RREDI detector: its 7-day rolling-median baseline sits higher than an Eckhardt baseflow, so the quick-flow volume above it is smaller, and the many additional small events lower the median. The runoff coefficient is not a predictor in the core models, so this does not affect the peak-shaving analysis; it does mean the coefficient is less comparable to the earlier run as an outcome.

5 Step 3 — physical wetland effectiveness W

The wetland metric follows $W_j = \sum_i A_i S_i K(T_{ij})$ with the connectivity term C dropped (it double-counts the travel-time kernel). Travel times T_{ij} come from a celerity field (channel speed = $\frac{5}{3} \times \text{Manning velocity}$) with wetland-roughness slowing overland flow; $S_i = M_{\text{type},i} \times d_i$ combines a Cowardin/HGM attenuation weight with a glaciated Volume–Area storage depth ($V = 0.01725A^{1.30}$), so $A_i S_i = M_{\text{type}} V_i$ is type-weighted wetland storage; $K(T) = e^{-T/\tau}$ with τ the catchment-median travel time. Table 3 gives the resulting gauge-level W and storage S_w (companion note documents the construction and literature in full).

Table 3: Step 3 — v2 wetland effectiveness and storage (contract schema).

Site	wet (km ²)	W (10 ⁶)	S_w (10 ⁶ m ³)	τ (h)
04087000	323.7	56.6	193.4	297
04086500	71.2	15.9	48.6	163
04086600	304.6	56.9	181.6	224
04087120	33.1	4.4	17.1	135
04087030	16.2	2.9	9.3	80
04087050	2.4	0.32	1.3	27
04087070	5.3	0.92	2.7	76
04087088	4.0	0.50	2.0	48
04087119	0.3	0.02	0.09	47

Caveat — travel-time calibration. Regressing observed RREDI hydrograph lag on the modeled travel time now yields a negative R^2 : RREDI’s short rising-limb lags (median 2–19 h) do not track the geometric basin travel time, so the model is retained with an adaptive (median- T) decay scale rather than a data-calibrated one. Absolute T is therefore still a relative index, as flagged previously.

6 Step 4 — controls

Per-gauge controls (drainage area, basin/channel slope, NLCD impervious interpolated to event year, POLARIS K_{sat} /sand, gNATSGO available water storage) are unchanged; impervious cover spans 4% (rural) to 53% (urban Honey Creek) and, as in v1, every control is collinear with the urban–rural gradient.

7 Step 5 — panel and collinearity

The stages join into 6,858 gauge-events, of which **5,907 are usable** ($P_e > 1$ mm and a finite runoff coefficient). Gauge-level predictor collinearity (Fig. 1) is essentially identical to v1 (wetland fraction vs impervious $r = -0.87$; W vs storage $r = 1.0$; W vs drainage area $r = 0.90$), because the gauges and geometry are unchanged. As before we use the intensive treatment $W_n = W/\text{area}$ and control drainage area explicitly.

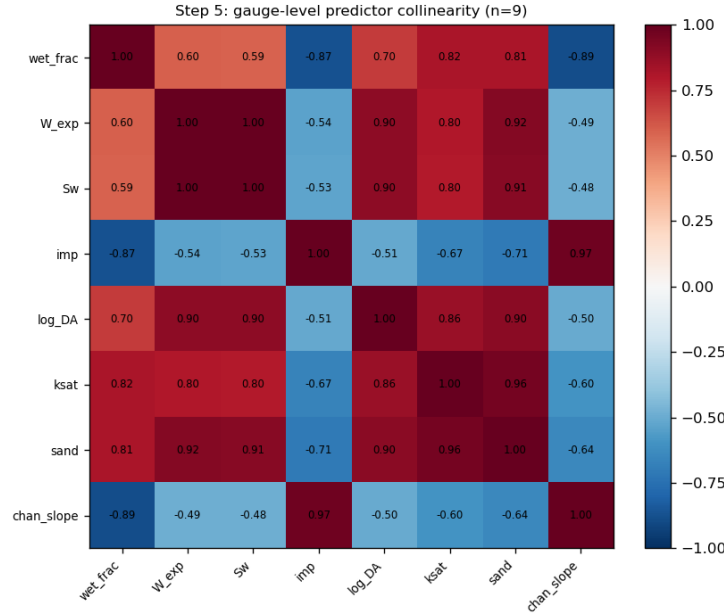


Figure 1: Gauge-level predictor collinearity ($n = 9$).

8 Step 6 — models and honest inference

Standardized predictors; nested Models 1–4 estimated as gauge random-intercept mixed models. For log peak discharge, rainfall dominates ($\hat{\beta}_P \approx 0.9$ –1.2), the wetland level effect is negative but confounded, and the peak-shaving interaction $W \cdot P$ is robustly negative (Table 4, Fig. 2).

Variance inflation confirms the identification structure: between-gauge terms are severely collinear (VIF: $S = 368$, $W = 268$, $P = 173$, impervious 87, channel slope 151), whereas the within-gauge interaction $W \cdot P$ has VIF 54. The placebo (permuting wetlands among gauges) cannot distinguish the observed W level effect from random relabeling ($p = 0.064$) but flags $W \cdot P$ as special ($p = 0.000$); leave-one-gauge-out gives a stable $W \cdot P$ (range $[-0.281, -0.211]$).

Table 4: Step 6 — MixedLM standardized coefficients, primary outcome $\log Q_p$ (p -values asymptotic; see wild bootstrap for honest inference).

Model	P	W	$W \cdot P$	V_e/S_w	$W \cdot (V_e/S_w)$
M1 basic	0.96	-0.29 (0.04)	—	—	—
M2 peak-shave	0.91	-0.13 (0.35)	-0.23 (0.00)	—	—
M3 saturation	1.22	-0.12 (0.18)	-0.23 (0.00)	-0.46 (0.00)	—
M4 nonlin. sat.	0.66	0.54 (0.34)	-0.55 (0.04)	0.36 (0.61)	0.46 (0.24)

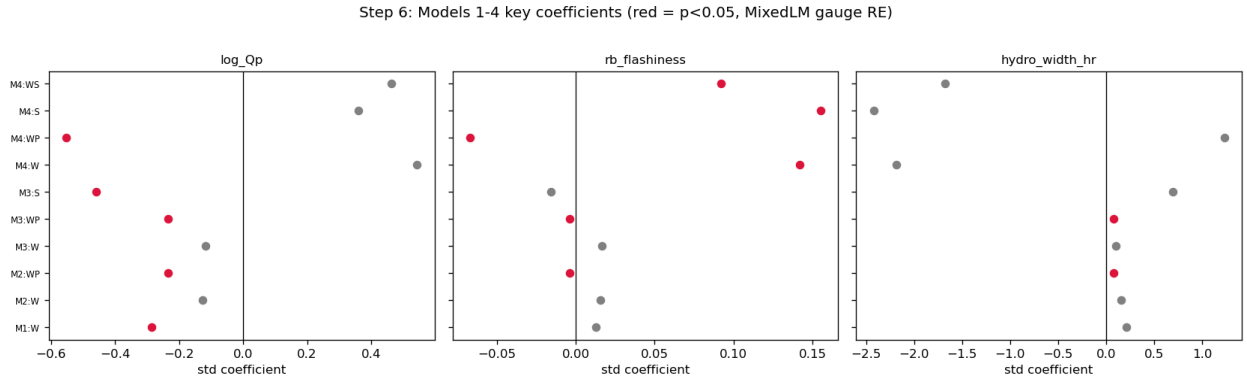


Figure 2: Models 1–4 key coefficients (MixedLM); red = asymptotic $p < 0.05$.

Wild cluster bootstrap. With nine clusters the asymptotic standard errors are unreliable, so we enumerate all $2^9 = 512$ Rademacher sign vectors exactly (Table 5). The peak-shaving interaction survives decisively on peak discharge ($W \cdot P = -0.233$, $p = 0.010$), and it is *larger* than the v1 estimate (-0.155 , $p = 0.029$): the upgraded events, rainfall, and wetland metric sharpen rather than overturn the finding. On the shape/timing outcomes the interaction is now only marginal ($p \approx 0.07$ – 0.08). The saturation term is nominally significant on peak discharge but has VIF = 368 (nearly collinear with rainfall P), so it is not cleanly interpretable and we do not read it as a saturation effect. The wetland level effect remains unidentified ($p = 0.084$).

9 Discussion

The v2 pipeline was a stress test of the central finding against three substantial, independent changes to the input variables: a different event detector, a $15\times$ finer rainfall product, and a physically rebuilt wetland metric. The peak-shaving mechanism —wetlands attenuating flood peaks disproportionately in larger storms— not only reproduces but strengthens (-0.233 vs -0.155) and remains robust under placebo, leave-one-gauge-out, and wild cluster bootstrap. That the result is invariant to, and sharpened by, these changes is strong evidence that it is a real signal rather than an artifact of any one construction. The shape/timing interactions weaken to marginal significance

Table 5: Wild cluster bootstrap (512 exact Rademacher, $G = 9$). p_{WCR} = restricted bootstrap- t ; CI = unrestricted percentile interval.

Outcome	Term	$\hat{\beta}$	cluster t	p_{WCR}	95% CI
$\log Q_p$	$W \cdot P$ (M2)	-0.233	-5.1	0.010	[-0.307, -0.158]
$\log Q_p$	V_e/S_w (M3) [†]	-0.337	-3.7	0.002	[-0.473, -0.200]
$\log Q_p$	W (M1)	-0.447	-9.5	0.084	[-0.533, -0.361]
$\log Q_p$	$W \cdot (V_e/S_w)$ (M4)	-0.102	-0.7	0.586	[-0.393, 0.188]
rb flashiness	$W \cdot P$ (M2)	-0.004	-3.0	0.068	[-0.007, -0.002]
hydro. width	$W \cdot P$ (M2)	0.089	2.9	0.082	[0.034, 0.143]

[†]Nominally significant but VIF= 368 (collinear with P); not interpreted as a saturation effect.

under the RREDI events, and the wetland *level* effect remains unidentifiable, exactly as the design predicts.

10 Limitations

(i) *Identification*: nine nested, urbanization-confounded gauges cap what is estimable; only the event-varying $W \cdot P$ is robust, and a causal level effect needs a regional independent sample. (ii) *Runoff coefficient* shifts with the RREDI baseline (not a core predictor). (iii) *Travel-time* absolute magnitudes are uncalibrated; W is a relative index. (iv) *Saturation* V_e/S_w is collinear with rainfall and not cleanly interpretable. (v) PRISM is finer in space but still daily in time. (vi) Associational, not causal.

11 Reproducibility and isolation

Everything runs inside `Mark_model_v2/`; no existing file is modified. Run order: `s1_build_events_flow.py` (RREDI) $\rightarrow$ `s2_build_events_rain.py` (PRISM) $\rightarrow$ `s3a_build_T_v2.py` $\rightarrow$ `s3b_build_W_v2.py` $\rightarrow$ `s3c_wetland_adapter.py` $\rightarrow$ `s4_build_controls.py` $\rightarrow$ `s5_build_panel.py` $\rightarrow$ `s6_fit_models.py` $\rightarrow$ `s6_wild_bootstrap.py`. Stages 4–7 are byte-identical to the original workflow; the only new code is the PRISM rainfall source, the wetland-schema adapter, and the `api_30_mm` alias. Public data sources: USGS NWIS, PRISM, USFWS NWI, NLCD, POLARIS, gNATSGO, USGS 3DEP.

Key references. Kohler & Linsley (1951, API); Lighthill & Whitham (1955, kinematic celerity); Kadlec (1990, wetland roughness); Wu & Lane (2016, glaciated Volume–Area); Acreman & Holden (2013, wetland type); Ameli & Creed (2019, wetland location); Cameron, Gelbach & Miller (2008, wild cluster bootstrap).